

10. Intro to Rademacher complexity (and mention VCdim)

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§6 in [SS] w, Rademacher Complexity taken from §3.1 in Mohri et al.

We've covered ① $|\mathcal{H}| < \infty$ (restrictive!) Doesn't cover finite dim. polynomials

② HW: axis-aligned rectangles, $|\mathcal{H}| = \infty$. $X = \mathbb{R}^d$, $\dim(\mathcal{H}) = 2d$ or a subset of a subspace

$|\mathcal{H}| = 1$
especially easy

Can we generalize?

We'll cover

① Rademacher Complexity (Mohri, and essentially used in later chapters of [SS])

Simple proofs, but computing R.C. may be impossible (eg, NP-Hard)
especially if $ERM_{\mathcal{H}}$ is difficult to compute

② Growth Function

③ VC-dimension, a way to bound the growth function, and
easier to compute or bound

④ Result: for binary classification, finite VC-dim is

necessary and sufficient for PAC learnability

"Fundamental Thm. of ML"

Rademacher Complexity (§3.1 Mohri, notation adapted a bit)

Will depend on $\mathcal{H}$ and loss function

$$\ell : \mathcal{H} \times \mathcal{Z} \rightarrow \mathbb{R} \text{ in [SS]}$$

$$\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R} \text{ in [Mohri et al.]}$$

$$\text{e.g. } \ell(h, (x, y)) = \ell(h(x), y)$$

we'll apply to a family of functions

$$\mathcal{F} = \{ f : (x, y) \mapsto \ell(h, (x, y)) \mid h \in \mathcal{H} \}$$

$$= \ell \circ \mathcal{H}$$

but it'll work for any family of functions $\mathcal{F}$, not just $\ell \circ \mathcal{H}$

$$\mathcal{F} \subseteq \mathbb{R}^{\mathcal{Z}} \quad \mathcal{Z} = \mathcal{X} \times \mathcal{Y}$$

Idea

Rademacher Complexity (RC) measures the richness / expressiveness

of $\mathcal{F}$ by measuring how well it can fit noise

(p. 2, Intro to Rademacher Complexity)

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Def Empirical Rademacher Complexity

$\mathcal{F}$ a family of fcn $f: \mathcal{Z} \rightarrow [a,b]$. Fix $S = (z_1, \dots, z_m)$
then empirical R.C. of $\mathcal{F}$ w.r.t. S is

$$\hat{R}_S(\mathcal{F}) = \mathbb{E}_{\sigma} \left[\sup_{f \in \mathcal{F}} \frac{1}{m} \sum_{i=1}^m \sigma_i \cdot f(z_i) \right]$$

Similar to "Gaussian width"
or "statistical dimension"

where $\sigma = (\sigma_1, \dots, \sigma_m)$, σ_i iid Rademacher variables

i.e. $\sigma_i = \begin{cases} +1 & \text{w.p. } .5 \\ -1 & \text{w.p. } .5 \end{cases}$ aka symmetric Bernoulli
or uniform on $\{-1, 1\}$

i.e., let $f_S := \begin{bmatrix} f(z_1) \\ \vdots \\ f(z_m) \end{bmatrix} \in \mathbb{R}^m$

then

$$\hat{R}_S(\mathcal{F}) = \frac{1}{m} \mathbb{E}_{\sigma} \left[\sup_{f \in \mathcal{F}} \langle \sigma, f_S \rangle \right]$$

DO NOT COMMUTE!!!!

See sub-additive note on last pg.

do as in-class exercise

i.e., best correlation w/ noise

Extremes: $\mathcal{F} = \{f\}$, $\hat{R}_S(\{f\}) = \frac{1}{m} \mathbb{E}_{\sigma} \langle \sigma, f_S \rangle = 0$. Best possible

vs. $\mathcal{F} = \text{all functions}$, say $f: \mathcal{Z} \rightarrow \{0,1\}$, then possible for some S Assume $z_i \neq z_j$ if $i \neq j$

for $\{f_S : f \in \mathcal{F}\} = \{0,1\}^m$ (If you use $\{0,1\}$ not $\{-1,1\}$, need to adjust by factor of 2)

Then $\sup_{f \in \mathcal{F}} \langle \sigma, f_S \rangle = m$, so $\hat{R}_S(\mathcal{F}) = \frac{1}{m} \mathbb{E}_{\sigma} m = 1$

Worst-possible
(if $[a,b] = [0,1]$)

Def Rademacher Complexity (not "empirical")

$$R_m(\mathcal{F}) := \mathbb{E}_{S \sim \mathcal{D}^m} \hat{R}_S(\mathcal{F})$$

Careful: [SS] uses different terminology:

we'll follow Mohri et al.

concept
 $\hat{R}_S(\mathcal{F})$
'Empirical R.C.'

concept
 $R_m(\mathcal{F}) = \mathbb{E}_{S \sim \mathcal{D}^m} \hat{R}_S(\mathcal{F})$
'R.C.'

Shalev-Shwartz + Ben-David

$R(\mathcal{F} \circ S)$
and $\mathcal{F} = \text{lost}$
'R.C.'

$\mathbb{E}_S R(\mathcal{F} \circ S)$
(no special notation)
'Expected R.C.'

(p. 3, Intro to Rademacher Complexity)

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How to use?

Recall for uniform convergence, S was " ϵ -representative" if

$$\sup_{h \in \mathcal{H}} |L_D(h) - \hat{L}_S(h)| \leq \epsilon$$

" $\forall h$ " same as " $\sup_h$ "

(which implied ERM worked: $L_D(\text{ERM}_{\mathcal{H}}(S)) \leq 2\epsilon + \min_{h \in \mathcal{H}} L_D(h)$)

Something very similar is the "representativeness" of S

(w.r.t. $\mathcal{H}, l$) as

$$\text{Rep}_D((\mathcal{H}, l), S) := \sup_{h \in \mathcal{H}} \underbrace{\mathbb{E}_{z \sim D} l(h, z)}_{L_D(h)} - \underbrace{\frac{1}{m} \sum_{i=1}^m l(h, z_i)}_{\hat{L}_S(h)}$$

Same thing, just no abs. value

or more generally

$$\underbrace{\Phi(S)}_{\text{in Mohri}} = \text{Rep}_D(\mathcal{F}, S) = \sup_{f \in \mathcal{F}} \underbrace{\mathbb{E}_{z \sim D} f(z)}_{\mathbb{E} f} - \underbrace{\frac{1}{m} \sum_{i=1}^m f(z_i)}_{\hat{\mathbb{E}}_S f}$$

Want this small.
clear

$$= \sup (\mathbb{E} f - \hat{\mathbb{E}}_S f) \text{ in shorthand notation.}$$

Intuitively, $2 \cdot \hat{\mathcal{R}}_S(\mathcal{F})$ is a reasonable estimate for $\text{Rep}_D(\mathcal{F}, S)$

Why?

in $\text{Rep}_D(\mathcal{F}, S)$ we have $\mathbb{E} f - \hat{\mathbb{E}}_S f$. Split $S = S_1 \cup S_2$ at random

estimate $\mathbb{E} f - \hat{\mathbb{E}}_S f$ by $\hat{\mathbb{E}}_{S_1} f - \hat{\mathbb{E}}_{S_2} f$

makes sense, but...
... Worried that f is a r.v.?
Good. But we'll take sup trick

Draw σ_i Rademacher

Let $S_1 = \{z \in [m] : \sigma_i = +1\}$, $S_2 = S \setminus S_1$, and suppose $|S_1| = m/2$ exactly

$$\text{then } \hat{\mathbb{E}}_{S_1} f - \hat{\mathbb{E}}_{S_2} f = \frac{1}{m/2} \sum_{i \in S_1} f(z_i) - \frac{1}{m/2} \sum_{i \in S_2} f(z_i)$$

$$= \frac{1}{m/2} \sum_i \sigma_i f(z_i)$$

Thus taking a $\sup_{f \in \mathcal{F}} (\dots)$, as we do in Rep_D and $\hat{\mathcal{R}}_S$,

$$\text{we get } \text{Rep}_D(\mathcal{F}, S) \approx 2 \cdot \hat{\mathcal{R}}_S(\mathcal{F})$$

(p. 4, Intro to Rademacher Complexity)

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Now, let's be slightly more careful and formalize the above:

Lemma 26.2 (Shalev-Shwartz) $\mathbb{E}_{S \sim \mathcal{D}^m} \text{Rep}_{\mathcal{D}}(F, S) \leq 2 \cdot \mathbb{E}_{S \sim \mathcal{D}^m} \hat{\mathcal{R}}_S(F)$
 $= 2 \cdot \mathcal{R}_m(F)$

Proof:

$$\begin{aligned} \mathbb{E}_{S \sim \mathcal{D}^m} \text{Rep}_{\mathcal{D}}(F, S) &:= \mathbb{E}_{S \sim \mathcal{D}^m} \sup_{f \in F} \mathbb{E} f - \mathbb{E}_S^\dagger f \\ &= \mathbb{E}_{S \sim \mathcal{D}^m} \sup_{f \in F} \underbrace{\mathbb{E}_{S' \sim \mathcal{D}^m}}_{\text{no effect}} \left(\underbrace{\mathbb{E}_{S'}^\dagger}_{\mathbb{E} f} f - \mathbb{E}_S^\dagger f \right) \end{aligned}$$

implicitly using Fubini
 $\mathbb{E}_{X,Y} = \mathbb{E}_X \mathbb{E}_Y = \mathbb{E}_Y \mathbb{E}_X$
 under mild assumptions.
 eg. in L^1 , $\mathbb{E} 1 \dots 1 < \infty$

True for us since $f: \mathcal{Z} \rightarrow [a,b]$
 is bounded,
 $\mathbb{E} |f| \leq c \cdot \|f\|_\infty$

$$\leq \mathbb{E}_{S, S'} \sup_{f \in F} \left(\mathbb{E}_{S'}^\dagger f - \mathbb{E}_S^\dagger f \right)$$

sup is sub-additive
 [see details later]

$$:= \mathbb{E}_{S, S'} \sup_{f \in F} \frac{1}{m} \sum_{i=1}^m f(z_i') - f(z_i)$$

$$= \mathbb{E}_{S, S'} \sup_{f \in F} \frac{1}{m} \sum_{i=1}^m \sigma_i (f(z_i') - f(z_i))$$

Symmetrization Argument

(maybe?)
 justified by
 sub-additive
 ↓

sketchy

$\sigma_i = 1$ OK

$\sigma_i = -1$ flips $z_i', z_i \dots$ but same distribution

for any σ_i

SUBTLE!

$$= \mathbb{E}_{S, S', \sigma} \sup_{f \in F} \frac{1}{m} \sum_{i=1}^m \sigma_i (f(z_i') - f(z_i))$$

true $\forall \sigma$ so true for $\mathbb{E}$

becomes + due to Rademacher

$$\leq \mathbb{E}_{S', \sigma} \sup_{f \in F} \frac{1}{m} \sum_{i=1}^m \sigma_i f(z_i') + \mathbb{E}_{S, \sigma} \sup_{f \in F} \frac{1}{m} \sum_{i=1}^m \sigma_i f(z_i)$$

$$\sup_f G(f) + H(f) \leq \sup_f G(f) + \sup_f H(f) \quad (\text{duh})$$

$$= 2 \mathbb{E}_{S, \sigma} \sup_{f \in F} \frac{1}{m} \sum_{i=1}^m \sigma_i f(z_i) = 2 \mathbb{E}_S \hat{\mathcal{R}}_S(F)$$

$$= 2 \mathcal{R}_m(F) \quad \square$$

sub-additive:

$$\forall a, b \quad g(a, b) \leq \sup_{b'} g(a, b')$$

$$\text{so } \forall b \quad \mathbb{E}_a g(a, b) \leq \mathbb{E}_a \sup_{b'} g(a, b')$$

so

$$\sup_b \mathbb{E}_a g(a, b) \leq \mathbb{E}_a \sup_{b'} g(a, b')$$

no $\leq$ possible

'26:

See "Commuting things" sheet

also true if replace $\mathbb{E}$ w/ infimum

Not true w/ strict inequalities
 $(g(a) < 0 \not\Rightarrow \sup g(a) < 0)$